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SUBJECT CODE	:	**MAT2001**
SUBJECT TITLE	:	**Statistics for Engineers**
LAB SLOT	:	**L 27+L28**
SEMESTER	:	**Winter Semester 2025-2026**

AIM:

1. Experience has shown that 20% of a manufactured product is of top quality. In one day's production of 400 articles, only 50 are of top quality. Write down the R programming code to test whether the production of the day chosen is a representative sample at 95% confidence level

KEY WORDS:

Example:

P = Population proportion
p = Sample proportion
n = Sample size
ZCal = Calculated Z value
ZTab = Table (critical) Z value
α (alpha) = Level of significance
H0 = Null hypothesis
H1 = Alternative hypothesis
SE = Standard Error

R-CODE:

```
print("H0 := p=P")
print("H1 := p!=P")

alpha=0.05
ZTab=qnorm(1-alpha/2)
ZTab

P=20/100
n=400
p=50/400

ZCal=(p-P)/sqrt(P*(1-P)/n)
ZCal

if(abs(ZCal)<abs(ZTab)) {
  print("H0 is accepted and H1 is rejected")
} else {
  print("H0 is rejected and H1 is accepted")
}
```

CONCLUSION:

i) **Population Proportion (P)** = 0.2

ii) **Sample Proportion (p)** = 0.125

iii) **Sample Size (n)** = 400

iv) **1. ZCal** = -3.75

v) **2. ZTab** = 1.96

vi) **3. Level of Significance (α)** = 0.05

Decision = H0 is rejected

Final Conclusion =At 95% confidence level, the sample is **not representative** of the population.

AIM:

2. Before an increase in excise duty on tea, 800 people out of a sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Write down the R programming code to test whether the significant decrease in the consumption of tea after the increase in duty at 1 % level of significance.

KEYWORDS:
p1 = Sample proportion (before duty)
p2 = Sample proportion (after duty)
n1 = Sample size (before duty)
n2 = Sample size (after duty)
P = Pooled proportion
ZCal = Calculated Z value
ZTab = Table (critical) Z value
α (alpha) = Level of significance
H0 = Null hypothesis
H1 = Alternative hypothesis
SE = Standard Error

R-Code:
```
print("H0 := p1 = p2")
print("H1 := p1 > p2")

alpha <- 0.01
ZTab <- qnorm(1 - alpha)
ZTab

n1 <- 1000
n2 <- 1200

p1 <- 800/1000
p2 <- 800/1200

P <- ((n1 * p1) + (n2 * p2)) / (n1 + n2)

ZCal <- (p1 - p2) / sqrt(P * (1 - P) * ((1/n1) + (1/n2)))
ZCal

if (abs(ZCal) < abs(ZTab)) {
```

```
 print("H0 is accepted and H1 is rejected")
} else {
 print("H0 is rejected and H1 is accepted")
}
```

CONCLUSION:

i) **p1 (Before duty)** = 0.8

ii) **p2 (After duty)** ≈ 0.6667

iii) **Sample Sizes** = n1 = 1000, n2 = 1200

iv) **ZCal** ≈ 7.02

v) **ZTab** ≈ 2.33

vi) **Level of Significance (α)** = 0.01

Decision = H0 is rejected

Final Conclusion =

At 1% level of significance, there is a **significant decrease in the consumption of tea after the increase in duty**

AIM:

3. A sample of 900 items is found to have a mean of 3.47 cm. Write down the R

programming code to test whether it can be reasonably regarded as a simple sample Mathematics– SAS– VIT from a population with mean 3.23 cm and SD 2.31 cm at 99% level of confidence.

KEYWORD:
μ (Mu) = Population mean
$\bar{x}$ (x0) = Sample mean
n = Sample size
σ (Sigma) = Population standard deviation
ZCal = Calculated Z value
ZTab = Table (critical) Z value
α (alpha) = Level of significance
H0 = Null hypothesis
H1 = Alternative hypothesis
SE = Standard Error

R-Code:

```
print("H0 := x0 = Mu")
print("H1 := x0 != Mu")

alpha <- 0.01
ZTab <- qnorm(1 - alpha/2)
ZTab

Mu <- 3.23
Sigma <- 2.31
n <- 900
x0 <- 3.47

ZCal <- (x0 - Mu) / (Sigma / sqrt(n))
ZCal

if (abs(ZCal) < abs(ZTab)) {
  print("H0 is accepted and H1 is rejected")
} else {
  print("H0 is rejected and H1 is accepted")
}
```

CONCLUSION:

i) **Sample Mean (x0)** = 3.47
ii) **Population Mean (μ)** = 3.23

iii) **Sample Size (n)** = 900

iv) **1. ZCal** ≈ 3.12

v) **2. ZTab** ≈ 2.576

vi) **3. Level of Significance (α)** = 0.01

4. **Decision** = H0 is rejected
5. **Final Conclusion** =
   At 99% confidence level, the sample **cannot be regarded as a simple sample** from the population.

AIM:
4. The average mark scored by 32 boys is 72 with a standard deviation of 8, while that for 36 girls is 70 with a standard deviation of 6. Write down the R programming code to test whether the boys are performing better than girls on the basis of average mark at 5 %level of significance.

KEYWORDS:

x̄1 = Mean of boys
x̄2 = Mean of girls
n1 = Number of boys
n2 = Number of girls
σ1 = SD of boys
σ2 = SD of girls
ZCal = Calculated Z value
ZTab = Table Z value
α (alpha) = Level of significance
H0 = Null hypothesis
H1 = Alternative hypothesis
SE = Standard Error

R-Code:

```
print("H0 := x1 = x2")

print("H1 := x1 > x2")

 alpha <- 0.05

ZTab <- qnorm(1 - alpha)

ZTab

 x1 <- 72

x2 <- 70

 s1 <- 8

s2 <- 6

 n1 <- 32

n2 <- 36

 ZCal <- (x1 - x2) / sqrt((s1^2 / n1) + (s2^2 / n2))

ZCal
```

```
 if (abs(ZCal) < abs(ZTab)) {
  print("H0 is accepted and H1 is rejected")
} else {
  print("H0 is rejected and H1 is accepted")
}
```

CONCLUSION:

i) **Mean of boys ($\bar{x}1$)** = 72

ii) **Mean of girls ($\bar{x}2$)** = 70

iii) **Sample Sizes** = n1 = 32, n2 = 36

iv) **1. ZCal** ≈ 1.17

v) **2. ZTab** ≈ 1.645

vi) **3. Level of Significance (α)** = 0.05

Decision = H0 is accepted
Final Conclusion =
At 5% level of significance, there is **no sufficient evidence to say boys perform better than girls**